

Class Basic Syntax

Basic syntax

Ek jaise bahut saare objects banane ke liye:

javascript

```
class User {  
  constructor(name) {  
    this.name = name;  
  }  
  sayHi() {  
    console.log(this.name);  
  }  
}  
  
let user = new User("John");  
user.sayHi();
```

- `new User(...)` par naya object banta hai aur `constructor` chalta hai.
- **Methods ke beech comma nahi lagta** (object literal se alag).

Class asal me kya hai?

Class ek **function** hi hai.

javascript

```
typeof User; // "function"  
User === User.prototype.constructor; // true
```

`class` likhne par JS:

1. `User` naam ka function banata hai (code constructor se).
2. Saare methods `User.prototype` me daal deta hai.

Sirf "syntactic sugar" nahi

Purane tareeke jaisa lagta hai, par kuch farak hain:

1. Class function bina `new` ke call nahi ho sakta (error).
2. Class ke methods **non-enumerable** hote hain (`for...in` me nahi aate).
3. Class ke andar ka code hamesha **strict mode** me chalta hai.

Class Expression

Function ki tarah class bhi expression ban sakti hai, aur naam de sakte ho jo sirf class ke andar dikhta hai.

javascript

```
let User = class MyClass { ... };
```

`makeClass(phrase)` jaisa function class return bhi kar sakta hai (on-demand classes).

Getters / Setters aur computed names

Object literal ki tarah:

javascript

```
class User {  
  constructor(name) { this.name = name; } // setter chalega  
  get name() { return this._name; }  
  set name(value) {  
    if (value.length < 4) return;  
    this._name = value;  
  }  
  ["say" + "Hi"]() { console.log("Hello"); } // computed name  
}
```

Ye getters/setters `User.prototype` me bante hain.

Class fields

Class me seedha property likh sakte ho:

javascript

```
class User {  
  name = "John";  
}
```

Ye **har object par** set hoti hai, `User.prototype` par nahi.

Fields se bound methods (this na khoye)

javascript

```
class Button {  
  constructor(value) { this.value = value; }  
  click = () => { console.log(this.value); }  
}  
setTimeout(new Button("hello").click, 1000); // hello
```

Arrow function field har object ka apna function banata hai jisme `this` sahi rehta hai. Event listeners me bahut kaam aata hai.

Yaad rakho

- Class = function + prototype methods.
- `new` ke bina call nahi hoti, strict mode me chalti hai.
- Fields object par set hoti hain, methods prototype par.